Title: A Cyber-Physical Threat Intelligence and Automated Incident Response Framework for Critical Cloud Infrastructure

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Proposal summary:

In late 2024 and around early 2025, there was a series of severed submarine cables in the Baltic and Red Seas that exposed a massive blind spot in modern cybersecurity. It showed us that our cloud relies entirely on fragile, physical choke points. Most Security Operation Centers currently monitor networks that are mostly logical. But when it comes to physical perimeter defense, it operates in total isolation, completely separate from the logical layer. So when I saw the viral open-source project God's Eye View, I thought of it as the perfect opportunity to bridge that link. While "God's Eye View" and other intelligence platforms can visualize real-time tracking data for military flights, seismic activities, and maritime vessels, they still remain just a passive tool that lacks any sort of automated incident response capabilities. My primary goal for this capstone project is to be able to develop and empirically validate the data with the Cyber-Physical Incident Response and a sort of Threat-Scoring Framework. I plan to also create a framework that would establish a model that would correlate dynamic geospatial threats with static digital infrastructure. The core objectives of this would include perhaps something such as a threat assessment taxonomy, while also designing a Python algorithm that would analyze suspected threats, and there would of course be an automated response mechanism that would alert the SOCs in charge and reroute network traffic before physical severance occurs.

Methodology to be used in the project:

I plan to experiment and test my program with an event simulation approach in a controlled environment. First, I will utilize the integrated vessel tracker in the application and use open source data feeds like Automatic Identification System (AIS) for the location, speed, course, identity, etc., of the vessels. Next, I will formulate the basic algorithmic core of the project by programming a Python backend for the feature that would calculate a normal Threat Score ranging from 0 to 100. I will test this algorithm thoroughly and make changes along the way as I observe its results. I will test the engine by injecting fake spoofed vessels that are meant to trigger an alert response. I will test various test scenarios with different ships, speeds, etc., to fine-tune the feature. These test scenarios will range from loitering vessels to vessels dropping anchor over subsea sables. If the calculated threat score breaches a predefined threshold, the engine will automatically trigger an alert and fire a webhook to simulate a pipeline.

Project expected outcomes:

I hope to provide a model that cybersecurity researchers and infrastructure operators can utilize to quantify cyber-physical risks. Along with the theoretical framework, I will deliver an open-source, functional, reproducible Python threat-hunting engine that would be capable of calculating different spatial threat scores while also dispatching alerts. The success of this system

will be backed by its empirical data, such as its validation metrics, which will be derived from the various tests and simulated scenarios. These tests will include charting detection accuracy, precision, recall, false-positive trade-offs, and mitigation latency. Ultimately, I am aiming to produce a fully fledged IEEE-formatted final paper that has the fully documented source code and perhaps a live demonstration that can showcase an automated traffic failover in response to a simulated vessel cable breach.